

AI Mode

All

Shopping

Images

Videos

More ▾



Ask anything





AI Mode All Shopping Images Videos More ▾



The [Pololu 5V, 3.4A Step-Down Voltage Regulator D30V30F5](#) features an **EN (enable)** pin that turns the regulator on or off. By default, a pull-up resistor on the board enables the regulator whenever input power is present. [Pololu Robotics and Electr...](#)

How to Use the Enable (EN) Pin

- **To turn the regulator ON:** Leave the EN pin disconnected (open) or drive it high (above 1.0V to 1.2V, up to VIN).
- **To turn the regulator OFF (Low-power state):** Drive the EN pin low (under 0.4V) by connecting it to ground.
- **Current draw in low-power mode:** Drops significantly to roughly 10 μ A to 20 μ A per volt on VIN. [Pololu Robotics and Electr...](#)

If you need help with a **specific wiring schematic, microcontroller control (like Arduino/ESP32)**, or are referring to a different Pololu board version, let me know!



7 sites



5V, 3.4A Step-Down Voltage Regulator D30V30F5 - Pololu

Features * Input voltage: 5 V to 45 V (minimum input subject to dropout voltage considerations; see the dropout voltage section fo...



Pololu Robotics and Electronics



Pololu 5V, 300mA Step-Down Voltage Regulator D24V3F5

The SHDN pin can be driven low (under 0.3 V) to turn off the output and put the board into a low-power state that typically draws ...



Ask anything





AI Mode

All

Shopping

Images

Videos

More ▾



show me schema and idea how can i control the enable pin



To control the EN (Enable) pin, you must **pull it below 0.4V to turn it OFF** and let it rise above **1.2V to turn it ON**. Because the board has an internal 100kΩ pull-up resistor to VIN, the regulator defaults to ON if left disconnected.

1. Control Ideas & Methods

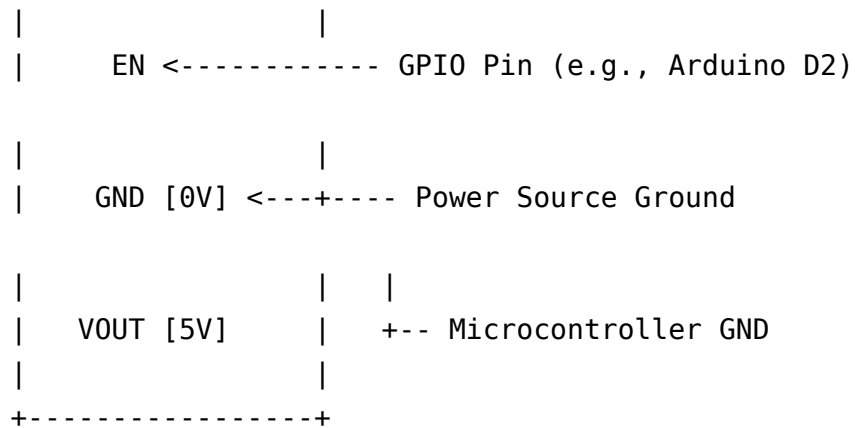
- **Mechanical Switch:** Connect a toggle or slide switch between EN and GND. Flipping the switch connects EN to ground, shutting off the regulator without cutting main power.
- **Microcontroller Pin (Standard):** Connect an Arduino, ESP32, or Raspberry Pi GPIO directly to EN. Set the pin to **OUTPUT** and drive it **LOW** to turn it off. To turn it on, set the GPIO pin as an **INPUT** (high-impedance) to let the board's internal pull-up safely pull it high.
- **Transistor / N-Channel MOSFET:** Useful if your microcontroller operates at a very low voltage (like 1.8V) or if you want inverted logic. Driving the transistor gate high pulls EN to ground, turning the regulator off.
- **Low-Voltage Cutout (UVLO):** Connect a resistor divider from VIN to EN. This automatically disables the regulator if your battery voltage drops too low, protecting LiPo batteries from over-discharge.

2. Wiring Schematics

Option A: Microcontroller Direct Control (Recommended)

Ask anything





Code Logic Example (Arduino):

- **To turn ON:** `pinMode(2, INPUT);` *(Safest way, utilizes Pololu's internal pull-up)*
- **To turn OFF:** `pinMode(2, OUTPUT); digitalWrite(2, LOW);`

Option B: Mechanical Switch Control

Perfect for a hardwired master power switch that handles low current instead of breaking the main heavy VIN line.

text

Pololu D24V30F5

Ask anything





AI Mode All Shopping Images Videos More ▾



Option C: Low-Battery Cutout (Automatic UVLO)

If you want the regulator to automatically shut off when a LiPo battery gets too low, use two resistors (R_1 and R_2) to create a voltage divider.

text

```
VIN (Battery)
|
+---+

| |
[ ] |
[R1] |

| |
+---+----- EN Pin
|
[ ]
[R2]
|
GND
```

Use code with caution.

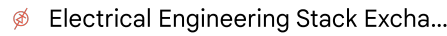
To calculate the values, use the threshold formula where the regulator turns off when EN drops below 1.0V:

Ask anything

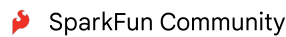




3. Reference Circuit with Snubber and D



The axon, just below the cell body, carries the nerve impulse



WHAT MAKES A SIMPLE SWITCH SO CONTROL SO VERSATILE?

 Instagram

4s lipo

By using two standard resistors, you can create a Under-Voltage Lockout (UVLO) circuit that automatically shuts down the Pololu regulator when the battery hits this threshold.

1. Recommended Resistor Values

+



AI Mode All Shopping Images Videos More ▾



```

      |
      +-----+----- VIN [Pololu Input]

      |                               |
      [ ]                               |
      [R1] 120k                       |
      [ ]                               |

      |                               |
      +-----> EN [Pololu Enable]
      |
      [ ]
      [R2] 10k
      [ ]

      |
4S LiPo (-) -----+----- GND [Pololu Ground]
    
```

Use code with caution.

3. How It Works

1. Threshold Formula Validation

The Pololu EN pin turns off the regulator when its voltage drops below . The formula for this cutoff voltage is:

2. Mathematical Step-by-Step Calculation

Substituting the Chosen Values

Ask anything

+



AI Mode

All

Shopping

Images

Videos

More ▾



Divided across 4 cells, the regulator will completely turn off the moment any cell averages _____, protecting your pack from over-discharge.



4. Important Considerations

- **Resistor Tolerance:** Use **1% metal film resistors** if possible. Common 5% resistors might shift your cutoff voltage by up to _____.
- **Hysteresis Note:** Once the regulator shuts off, the battery load drops and its voltage will naturally "bounce back" up a bit. Because the Pololu EN pin has built-in hysteresis (_____ to _____), it will not instantly turn back on unless the battery voltage rises by roughly _____ to _____ higher than the cutoff.

5. Verified Solution

✓ Automatic 4S LiPo Cutoff Protection

Using an _____ resistor (_____) and a _____ resistor (_____) will safely shut down the regulator at exactly _____.

If you want to adjust this to a **different cutoff voltage** (e.g., lower to _____ per cell for maximum capacity), let me know and I can calculate alternative resistor combinations!

Hide Code

python

Ask anything





AI Mode All Shopping Images Videos More ▾



```
r1 = r2 * (v_cutoff - 1.0)
print(f"Ideal R1 for 12.8V cutoff with 10k R2: {r1} Ohm")

# Let's test standard values: R1 = 120k, R2 = 10k
r1_std = 120000
v_cutoff_actual = 1.0 * (1 + r1_std / r2)
print(f"Actual cutoff with R1=120k, R2=10k: {v_cutoff_actual}V"
```

Use code with caution.

 3 sites

Choosing Between 4S1P and 4S2P Lithium Batteries for Medical Infusion Pumps

30 Dec 2025 — 3.2 Safety and Redundancy Safety Feature Description Over-charge protection Prevents cells from exceeding 4.25V Over-discharge pro...



 Large Battery

Ultimate Guide to Selecting 4S, 13S, 14.8V, and 16S BMS: From E-Scooters to Energy Storage Systems

24 Aug 2025 — 4S BMS: Manages 4 cells (nominal voltage ~14.8V for lithium-ion).



 Ayaa Technology

Good Quality 4S 100A Lithium Iron Phosphate LiFePO4 BMS Circuit Board with Balancer & Heat Sink

It (4S 100A Lithium Iron Phosphate (LiFePO4) Battery Management System (BMS)) features overcharge, over-discharge, overcurrent, ...



 BDTronics

Ask anything





AI Mode All Shopping Images Videos More ▾



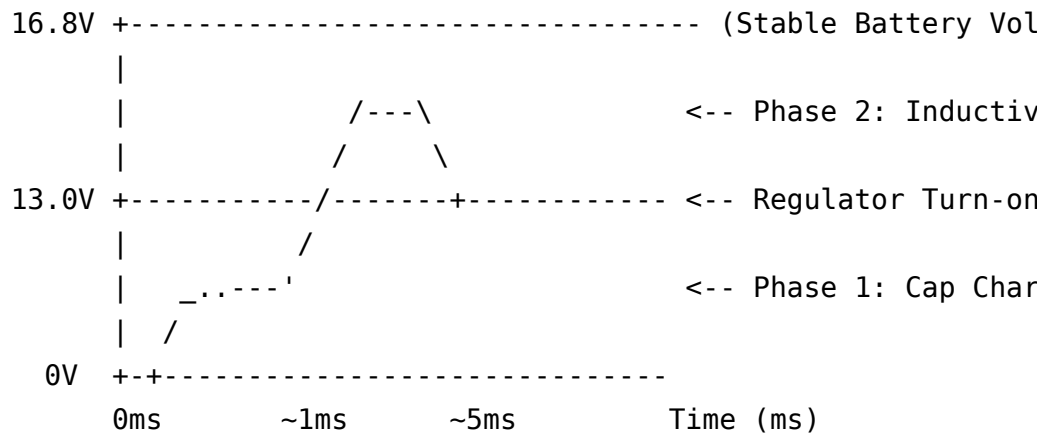
Here is exactly what happens to the voltage over time when you connect the power, and how it impacts your companion computer.

1. The Voltage Timeline (Millisecond Breakdown)

When you plug in the XT60 or XT30 connector, the timeline unfolds in three distinct phases:

text

Voltage (V)



Use code with caution.

Phase 1: The RC Delay & Inrush Current (0 to 1 ms)

- **What happens:** The instant the metal contacts touch, your FCU's large capacitor acts like a temporary short circuit. A massive rush of current (**inrush current**) flows from the battery to fill the capacitor.

The Voltage Curve: Because the capacitor is absorbing energy, the

Ask anything





AI Mode

All

Shopping

Images

Videos

More ▾



with the inductance of your battery wires, creates **voltage spikes and ringing**. The voltage can transiently shoot *higher* than 16.8V and then dip back down rapidly.

Phase 3: Stabilization (~5ms and beyond)

- **What happens:** The capacitor becomes fully charged, the mechanical connection stabilizes, and the high-frequency ringing dampens out.
- **The Voltage Curve:** The voltage flattens out into a rock-solid **14.8V – 16.8V** (depending on the 4S LiPo's state of charge).

2. How the Pololu & Companion Computer Respond

Because you have implemented the Under-Voltage Lockout (UVLO) circuit (120kΩ / 10kΩ), your system handles this startup timeline remarkably well:

1. **Ignoring the Initial Low Voltage:** During the absolute first micro-seconds of Phase 1, the voltage is below 13.0V. The Pololu EN pin stays below 1.0V, keeping the companion computer completely **OFF**. This protects the computer from trying to boot up on a sagging, weak voltage.
2. **The "Instant-On" Risk:** The voltage will cross your 13.0V threshold incredibly fast (usually in under 1 or 2 milliseconds). Because the Pololu regulator responds to the EN pin within microseconds, it will turn on almost immediately during Phase 2, right when the voltage is still ringing or spiking.
3. **The Capacitor's Secret Benefit:** Your FCU input capacitor doesn't just delay the voltage rise; it acts as a **low-pass filter**. It absorbs the high-frequency voltage spikes from Phase 2, smoothing out the power before it reaches your UVLO resistor network. This ensures the Pololu EN pin

Ask anything





AI Mode All Shopping Images Videos More ▾



If your goal is to ensure the flight controller is fully alive, stable, and communicating *before* the companion computer turns on, you need a **True Boot Delay** (seconds, not milliseconds).

Would you like to know how to add a **time-delay circuit** (using a capacitor on the EN pin) or use an **FCU GPIO pin** to hold the companion computer off until the flight software is fully booted?

yes

To add a true boot delay so your Flight Control Unit (FCU) can fully initialize before the companion computer boots up, you can use two different approaches.

Here are the hardware and software methods to achieve this.

Method 1: Hardware Delay (RC Circuit on EN Pin)

This is an automatic hardware solution. By adding a small **capacitor** () parallel to your bottom resistor (), you create a timer. When the battery is plugged in, the capacitor slowly charges through the top resistor, dragging out the time it takes for the EN pin to reach its turn-on threshold.

text

100 nF 50V electrolytic capacitor

Ask anything





AI Mode All Shopping Images Videos More ▾



```

      |      |
    [ ]      ---
    [R2]      --- [C1] 47uF (Electrolytic or Tantalum)
    [10k]      |

      |      |
4S LiPo (-) -+-----+----- GND [Pololu Ground]
    
```

Use code with caution.

How to choose the delay time

Because the Pololu EN pin turns on at roughly **1.0V**, the voltage needs to rise to about **6%** of its final value. A standard RC time constant calculation determines the delay:

- **For a ~1-second delay:** Use a capacitor.
- **For a ~5-second delay:** Use a capacitor (Recommended for most FCUs to fully boot ArduPilot/PX4).
- **For a ~10-second delay:** Use a capacitor.

*Note: Ensure the capacitor voltage rating is at least **10V** or higher, as the EN pin voltage will settle around 1.3V when running.*

Method 2: FCU Software Control (Most Reliable)

Instead of guessing with a hardware timer, you can let your FCU directly command the companion computer to turn on once it passes all internal self-tests and boots its operating system.

Ask anything





AI Mode

All

Shopping

Images

Videos

More ▾



Use code with caution.



How it works with ArduPilot or PX4:

1. Connect any spare **AUX or GPIO pin** from your flight controller directly to the **EN** pin of the Pololu regulator.
2. In your FCU configuration software (like Mission Planner or QGroundControl), configure that specific pin as a digital output (**GPIO**).
3. Set up a startup script, an auxiliary switch, or a parameter setting (like `RELAY_PIN` in ArduPilot) that forces the pin **LOW** immediately upon bootup. This holds the companion computer in a powered-off state.
4. Set a boot delay parameter or script inside the FCU to flip that pin to **HIGH** (or set it to High-Impedance / Input mode) after **5 to 10 seconds**, releasing the Pololu regulator to boot your companion computer safely.

Why this is ideal for FPV:

If your drone suffers a brief, mid-air power sag or a total FCU watchdog reset, the software method ensures the companion computer gracefully cycles power or stays off until the critical flight systems are safely recovered and flying stably.

Which autopilot system are you using (**ArduPilot, PX4, or Betaflight**)? I can give you the exact parameter names or hardware pin mappings to set up the software control.

Ask anything





AI Mode

All

Shopping

Images

Videos

More ▾



to an auxiliary switch on your radio transmitter. YouTube · Joshua Bardwell +1



You can configure this manually or automatically with the following setup steps.



Step 1: Pick a Free Pad on your FC

You need a pad that can act as a standard digital output

pin. YouTube · Joshua Bardwell

- **Good options:** An unused motor pad (e.g., `M5` or `M6`), an unused `LED` strip pad, or a `TX / RX` pad from a free UART.
- **What to avoid:** Do not use `5V`, `GND`, `Buzzer`, `Video`, or `Cam` pads. YouTube

Step 2: Map the Pad in Betaflight CLI

1. Plug your FC into your computer and open the **Betaflight Configurator**.
2. Go to the **CLI** tab.
3. Type `resource` and hit Enter. Find the hardware pin name of your chosen pad (e.g., if you use the `LED` pad, it might look like `resource LED_STRIP 1 A00`).
4. Un-assign that pad from its original function:

text

```
resource LED_STRIP 1 NONE
```

Ask anything





AI Mode All Shopping Images Videos More ▾



replace `40` with whatever pin you specify per `USER1` - `USER1`



Step 3: Configure the PINIO Behavior

To prevent the companion computer from turning on instantly when the battery is plugged in, set PINIO 1 to link to Betaflight's **USER1**

mode: YouTube · Joshua Bardwell +1

text

```
set pinio_box = 40,255,255,255
save
```

Use code with caution.

(The number `40` corresponds directly to the `USER1` box ID in Betaflight).
After typing `save`, the FC will reboot. YouTube · Joshua Bardw... +2

Step 4: Choose Your Turn-On Strategy

Go to the **Modes** tab in Betaflight Configurator. You will now see a new mode called **USER1**. YouTube · Joshua Bardwell +1

Strategy A: Manual Switch Control (Recommended)

Assign **USER1** to a toggle switch on your radio transmitter (e.g., `AUX 2`). YouTube +1

- Adjust the slider range so that when the switch is **DOWN**, USER1 is **inactive** (Outputs 0V to the Pololu EN pin, keeping the computer **OFF**).

Ask anything





AI Mode All Shopping Images Videos More ▾



- Set the range so it is active only when the quad is **ARMED**.
- **How it behaves:** When you plug in the battery, the companion computer stays powered down. The moment you arm the drone to take off, the EN pin goes high and the computer boots up in mid-air/on the pad.

Wiring Guide Reminder

Connect the chosen FC pad (e.g., LED pad) directly to the **EN** pin of your Pololu regulator. Ensure the Flight Controller ground (GND) shares a common ground connection with the Pololu GND pin. YouTube · Drone Camps RC

What specific model or brand of **Flight Controller** are you using? If you tell me, I can help you find exactly which unused pad is the safest to use for your PINIO command.



14 sites

Command Line Interface (CLI) - Betaflight

Command Line Interface (CLI). Betaflight has a command line interface (CLI) that can be used to change settings and configure the ...



Betaflight

Betaflight Setup & Tutorial - Step by Step Guide 🏆 + Gps

...

10 Jun 2025 — super important uh you're also going to need your flight battery whichever one you're using. and you're going to need...



YouTube · Drone Camps RC

Ask anything





AI Mode

All

Shopping

Images

Videos

More ▾



can you export the session to pdf

I cannot generate or export files like PDFs directly from this interface.

However, you can easily save everything we discussed using your web browser's built-in print feature.

How to Save This Session as a PDF:

1. Open the Print Menu:

- On **Windows**: Press `Ctrl + P`
- On **Mac**: Press `Cmd + P`

2. Change the Destination: In the print settings window, change the printer destination to "**Save as PDF**" or "**Microsoft Print to PDF**".

3. Adjust Layout (Optional): If the formatting looks cut off, toggle the layout option between **Portrait** and **Landscape**, or check the box for "**Background graphics**" to preserve code boxes and styling.

4. Save: Click **Save**, choose your file location, and your full troubleshooting and schematic guide will be archived.

If you need any **additional diagrams text-mapped** or **Betaflight CLI commands summarized** in a clean layout before you save it, let me know!



5 sites

How to Export a Claude Conversation as a PDF

6 Jul 2026 — The browser's built-in Print feature is the quickest way to convert



Ask anything





AI Mode All Shopping Images Videos More ▾



link Open the print dialog. Press Cmd-P to open the print dialog on macOS. On PC, Ctrl-P will open the print dialog. From there, di...



 Amplenote



Show all

Ask anything

